3. Verify the following:

(i) $[(-2)+(-9)]+6=(-2)+[(-9)+6]$ (ii) $[7+(-8)]+(-5)=7+[(-8)+(-5)]$

(iii) $[(-11)+5]+(-14)=(-11)+[5+(-14)]$

(iv) $(-5)+[(-32)+(-2)]=[(-5)+(-32)]+(-2)$

4. Find the missing integers:

(i) $0+(-95)=$ _____ (ii) $-611+$ _____ $=-611$

(iii) _____$+0=$_____ (iv) $0+(-140)=$ _____

5. Complete the following:

(i) $-603+603=$ ___ (ii) $9847+(-9847)=$ ___ (iii) $1652+$ ___ $=0$

(iv) $-777+$ ___ $=0$ (v) ___$+5281=0$

Example 1.7 (i) Are 120 + 51 and 51 + 120 equal?

(ii) Are $(-5)+[(-4)+(-3)]$ and $[(-5)+(-4)]+(-3)$ equal?

Solution

(i) When we add, 120 + 51 = 171 ; 51 + 120 = 171

In both the cases we get same answer. This means that integers can be added in any order. Hence, addition of integers is commutative.

(ii) $(-5)+[(-4)+(-3)]$ and $[(-5)+(-4)]+(-3)$

In $(-5)+[(-4)+(-3)]$, (-4) and (-3) are added first and their result is then added with (-5).

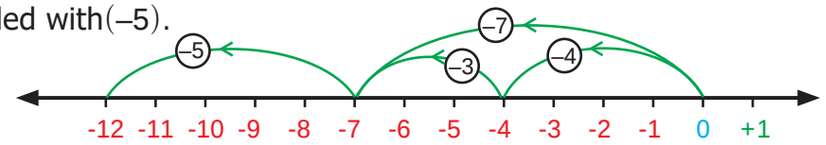


Fig. 1.11

$(-5)+[(-4)+(-3)]=-12$

Whereas in $[(-5)+(-4)]+(-3)$, (-4) and (-3) are added first and then the result is added with (-5)

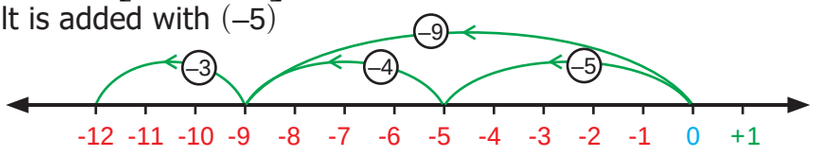


Fig. 1.12

$[(-5)+(-4)]+(-3)=-12$

In both the cases, we get -12

That is $(-5)+[(-4)+(-3)]=[(-5)+(-4)]+(-3)$

So, addition is associative.

Example 1.8 Find the missing integers (i) $0+(-2345)=$ _____ (ii) $23479+$ _____ $=0$

Solution (i) $0+(-2345)=-2345$

(ii) $23479+(-23479)=0$

Therefore, additive inverse of 23479 is –23479

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